Exit Ticket

Module 2, Topic 2, Lessons 1 and 2: Vertical and Horizontal Transformations · IEP Version

Name _____ Date _____ Period _____

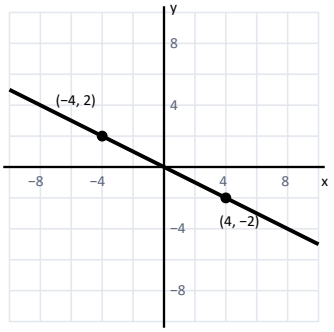
Reminders

- **Outside** the parentheses changes the outputs (up/down): f(x) + d moves up d, f(x) − d moves down d. a · f(x) stretches if a > 1, compresses if a is between 0 and 1, and reflects (flips) if a is negative.
- **Inside** the parentheses changes the inputs (left/right) and works the **opposite** way: f(x − h) moves right h, f(x + h) moves left h. f(bx) squeezes by 1/b.
- Slope = rise ÷ run. Falls left to right → negative slope.

1. The graph of linear function m is shown on the grid. Given f(x) = x and m(x) = a · f(x), what is the value of a?

Count from (−4, 2) to (4, −2): rise = ____ , run = ____ . Is the line going up or down?

a =



2. Maya graphed f(x) and g(x) = f(x + 6) on the same coordinate grid. Which statement describes how the graphs of f and g are related?

Is the + 6 inside or outside the parentheses?

A. The graph of f(x) is translated up 6 units to produce g(x).

B. The graph of f(x) is translated right 6 units to produce g(x).

C. The graph of f(x) is translated left 6 units to produce g(x).

3. What is the y-intercept of the linear function created by translating f(x), represented in the table, 3 units left?

Every point moves 3 left. Which x-value lands on x = 0? x = ____ , so f(____) = ____ .

y-intercept =

x	f(x)
−1	−1
0	1
1	3
2	5
3	7

4. f(x) = x and g(x) = f(3x). Graph g(x) on the grid and write its equation.

g(1) = f(3 · 1) = ____ g(2) = f(3 · 2) = ____ Plot (1, __) and (2, __).

Equation of g(x):

